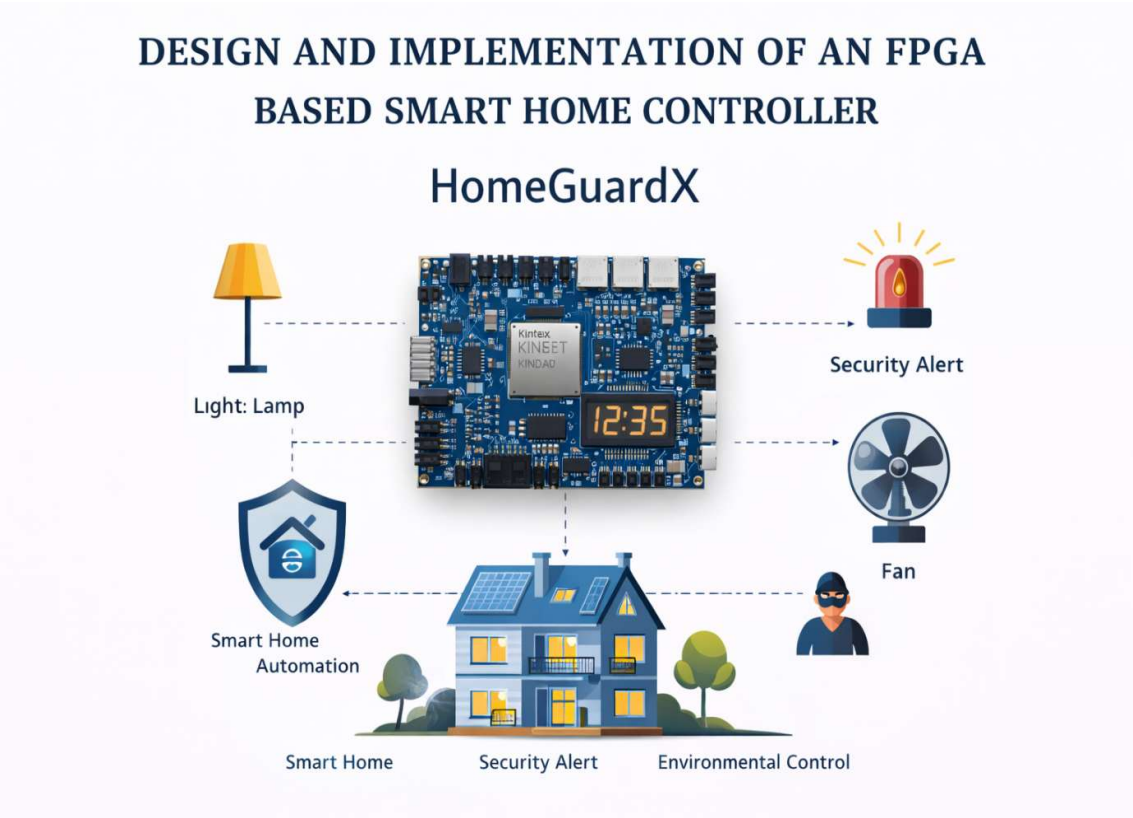
Design and Implementation of an FPGA Based Smart Home Controller - HomeGuardX



Submitted by:

Prakruthi D

Table of Contents

Table of Figures ---------- 3
1. Introduction ---------- 5
2. Problem Definition ---------- 6
3. Objectives ---------- 6
4. System Architecture Overview ---------- 7
5. Module Description ---------- 9
6. Simulation Results ---------- 13
7. FPGA Implementation ---------- 16
8. Resource Utilization and Timing Analysis ---------- 18
9. Conclusion ---------- 18
10. Future Scope ---------- 18

Table of Figures

Figure 1 System Architecture 7
Figure 2 RTL Schematic 8
Figure 3 Block diagram of the Configuration Registers module 9
Figure 4 Block diagram of the Temperature Fan Controller . 10
Figure 5 Block diagram of the Light Controller 10
Figure 6 Block diagram of the Intruder Alert Controller 11
Figure 7 Block diagram of the Timer Counter module 12
Figure 8 Block diagram of the Top-Level Controller integrating all submodules 12
Figure 9 Timing diagram of the temperature fan controller. . 13
Figure 10 Timing diagram of the light controller..... 14
Figure 11 Timing diagram of the Intruder Alert Controller showing alert activation 14
Figure 12 Behavioral simulation waveform of the top-level controller 15
Figure 13 FPGA resource utilization summary..... 16
Figure 14 Timing analysis summary after synthesis 16
Figure 15 Successful bitstream generation in Vivado 17
Figure 16 FPGA pin mapping defined using Xilinx Design Constraints 17

ABSTRACT

Smart home automation has become an important application in modern embedded systems, as it helps in improving comfort, safety, and efficient energy usage. This project focuses on the design and implementation of an FPGA-based Smart Home Controller using Verilog HDL.

The system consists of multiple functional modules including a temperature-based fan controller, an automatic light controller, and an intruder alert system. Configurable threshold values are stored using configuration registers, allowing flexibility in system operation. All modules are integrated using a top-level controller to form a complete system.

The design is verified using behavioural simulation, and FPGA synthesizability is confirmed through synthesis, implementation, and successful bitstream generation using Xilinx Vivado targeting an Artix-7 FPGA. Simulation waveforms, block diagrams, FSM diagrams, timing analysis, and resource utilization reports are used to validate the correctness and efficiency of the proposed system.

1. Introduction

1.1 General Introduction

With the rapid growth of technology, home automation systems have gained significant importance. Smart homes aim to automate everyday activities such as controlling lighting, regulating temperature, and providing security alerts without continuous human intervention. Traditional smart home systems are usually implemented using microcontrollers, which execute tasks sequentially and may face limitations in real-time applications.

Field Programmable Gate Arrays (FPGAs) offer an efficient alternative due to their parallel processing capability, high speed, and reconfigurability. FPGA-based systems are well suited for applications that require real-time response and reliable operation.

1.2 Motivation

The motivation for this project is to understand and apply FPGA-based design concepts to a real-world application. By implementing a smart home controller on an FPGA, the project demonstrates how multiple tasks can be handled simultaneously using hardware logic. This project also provides hands-on experience in Verilog HDL coding, simulation, synthesis, and FPGA implementation.

1.3 Project Objectives

The main objectives of this project are:

- To design a smart home controller using Verilog HDL
- To implement temperature-based fan control
- To implement automatic lighting control based on ambient light
- To design an intruder alert system using FSM
- To integrate all modules using a top-level controller
- To verify functionality using simulation and FPGA synthesis

2. Problem Definition

Modern homes require intelligent systems that can automatically respond to environmental conditions and security threats. Manual control of appliances often leads to inefficient energy usage and delayed response during critical situations.

The problem addressed in this project is the design of a hardware-based smart home controller that can:

- Monitor temperature and control a cooling fan
- Automatically switch room lights based on ambient light levels
- Detect intrusions and trigger security alerts
- Allow configurable threshold values
- Operate reliably in real time

The system should be modular, scalable, and suitable for FPGA implementation.

3. Objectives

The key objectives of the project are:

- To design individual functional modules for temperature control, lighting control, and intrusion detection
- To implement configuration registers for programmable threshold values
- To integrate all modules using a top-level controller
- To verify functionality using behavioral simulations
- To analyze FPGA resource utilization and timing performance

4. System Architecture Overview

The Smart Home Controller is designed using a modular and hierarchical architecture. Each functional unit is implemented as a separate module, and all modules are connected using a top-level controller.

The major modules in the system are:

- Configuration Registers
- Temperature Fan Controller
- Light Controller
- Intruder Alert Controller
- Timer Counter
- Top-Level Controller

This modular approach improves readability, debugging, and future scalability of the design.

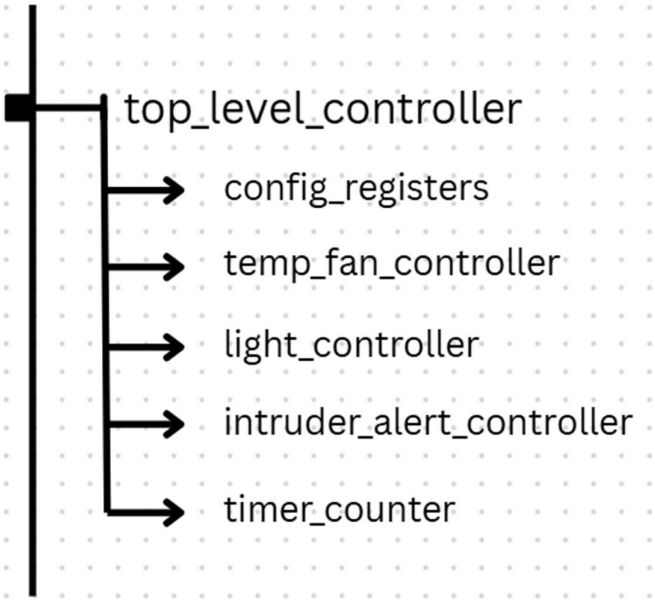


Figure 1 System Architecture

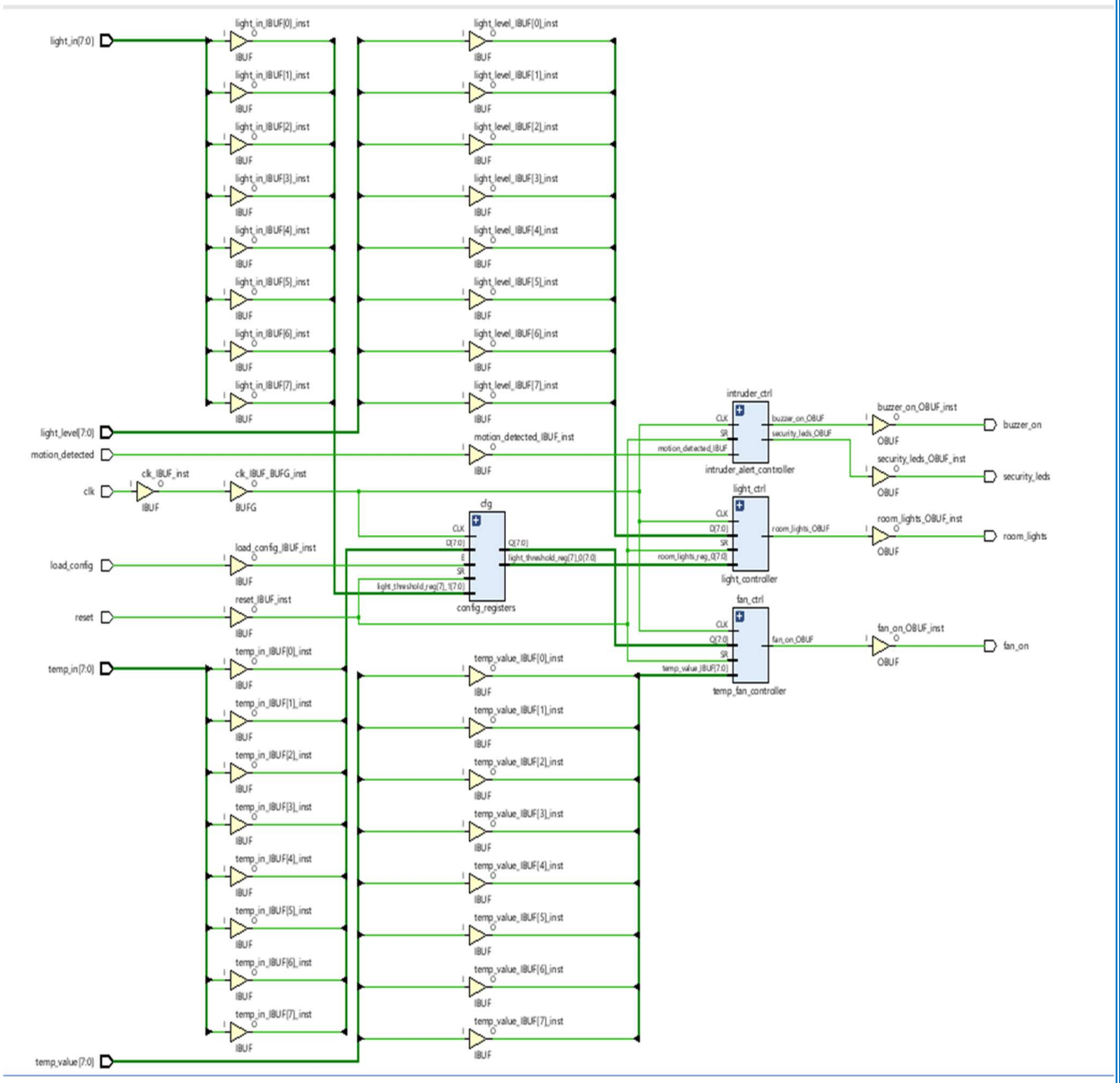


Figure 2 RTL Schematic

5. Module Description

5.1 Configuration Registers

The configuration registers module stores threshold values for temperature and light intensity. These values can be updated using a load control signal. The stored thresholds are provided to the respective control modules.

This module allows flexibility in system operation without modifying the HDL code.

Figure 3 Block diagram of the Configuration Registers module

5.2 Temperature-Based Fan Controller

The temperature fan controller compares the current temperature value with a predefined threshold.

- If the temperature exceeds the threshold, the fan is turned ON
- If the temperature is below the threshold, the fan remains OFF

This module uses simple comparison logic and does not require an FSM.

Figure 4 Block diagram of the Temperature Fan Controller

5.3 Light Controller

The light controller monitors the ambient light level and compares it with the light threshold value.

- When ambient light is low, room lights are turned ON
- When sufficient light is available, room lights are turned OFF

This helps in conserving energy and improves automation.

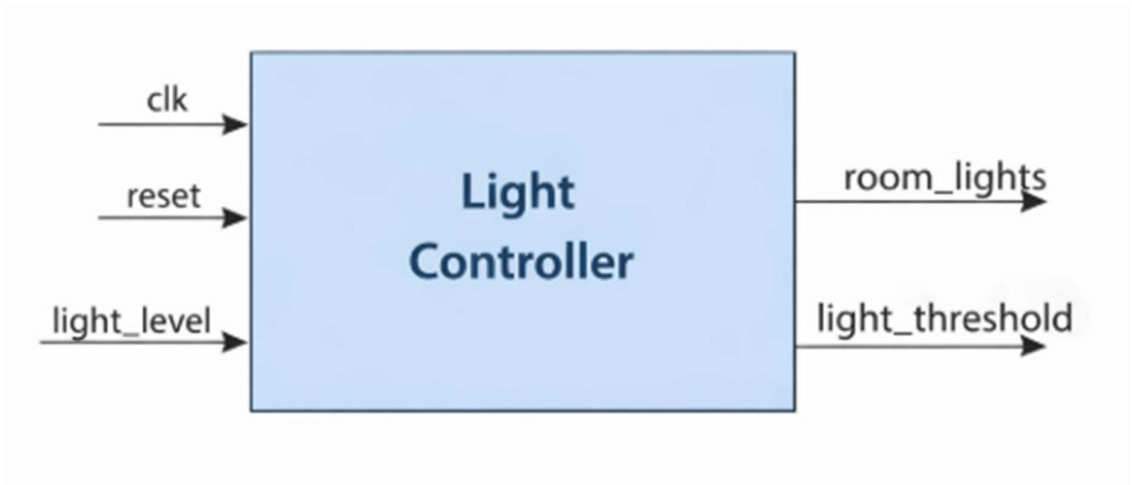


Figure 5 Block diagram of the Light Controller

5.4 Intruder Alert Controller

The intruder alert controller detects motion and activates security alerts. This module is implemented using a Finite State Machine (FSM) to ensure reliable operation.

When motion is detected:

- Security LEDs are turned ON
- Buzzer is activated
- Alert remains active for a predefined duration

Figure 6 Block diagram of the Intruder Alert Controller

5.5 Timer Counter

The timer counter module generates timing signals used for alert duration and optional blinking functionality. It supports time-based operations in the intruder alert system.

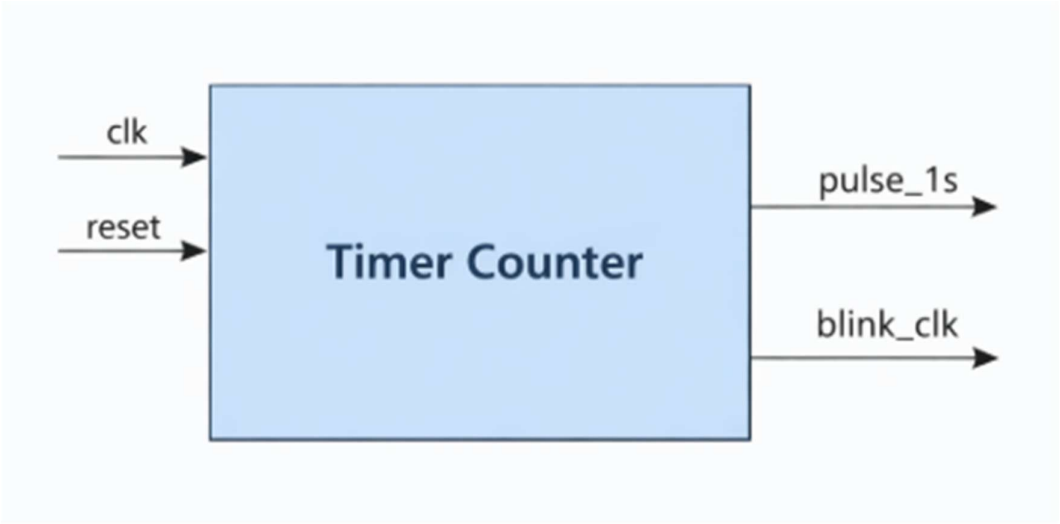


Figure 7 Block diagram of the Timer Counter module

5.6 Top-Level Controller

The top-level controller integrates all submodules. It distributes clock and reset signals, routes configuration data, and collects outputs from individual controllers to generate final system outputs.

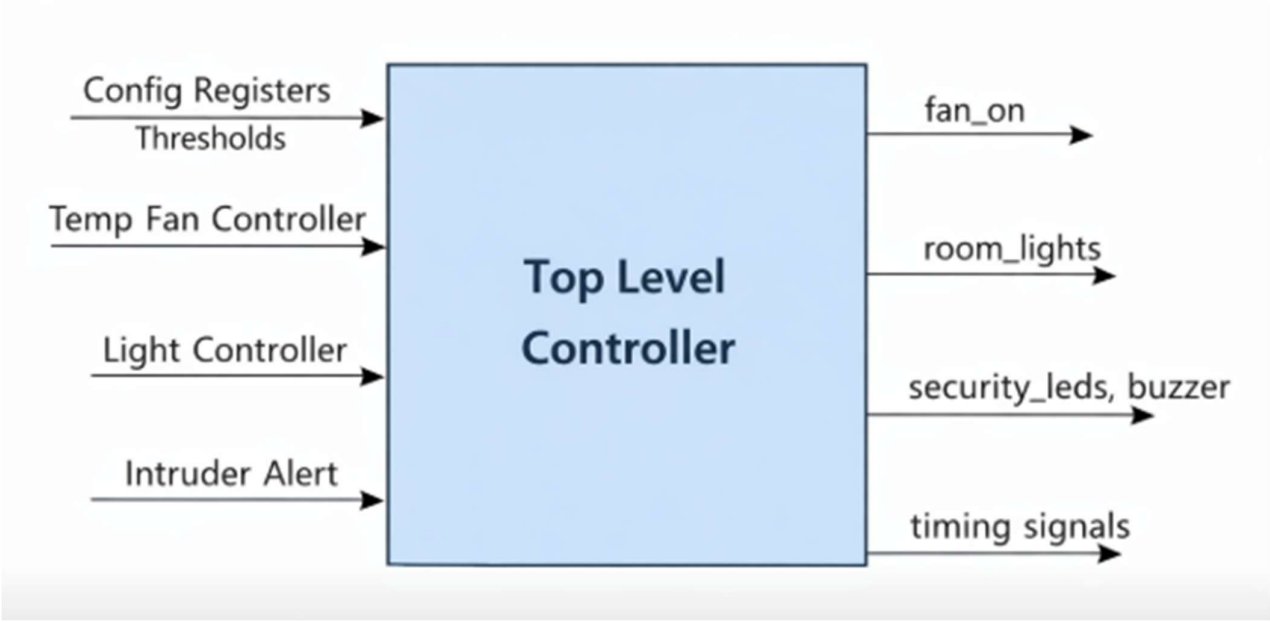


Figure 8 Block diagram of the Top-Level Controller integrating all submodules

6. Simulation Results

Behavioural simulations were performed for each major module using Verilog testbenches in Xilinx Vivado. Simulation waveforms confirm the correct operation of all modules under different input conditions.

- Temperature fan controller simulation verifies fan activation above threshold.

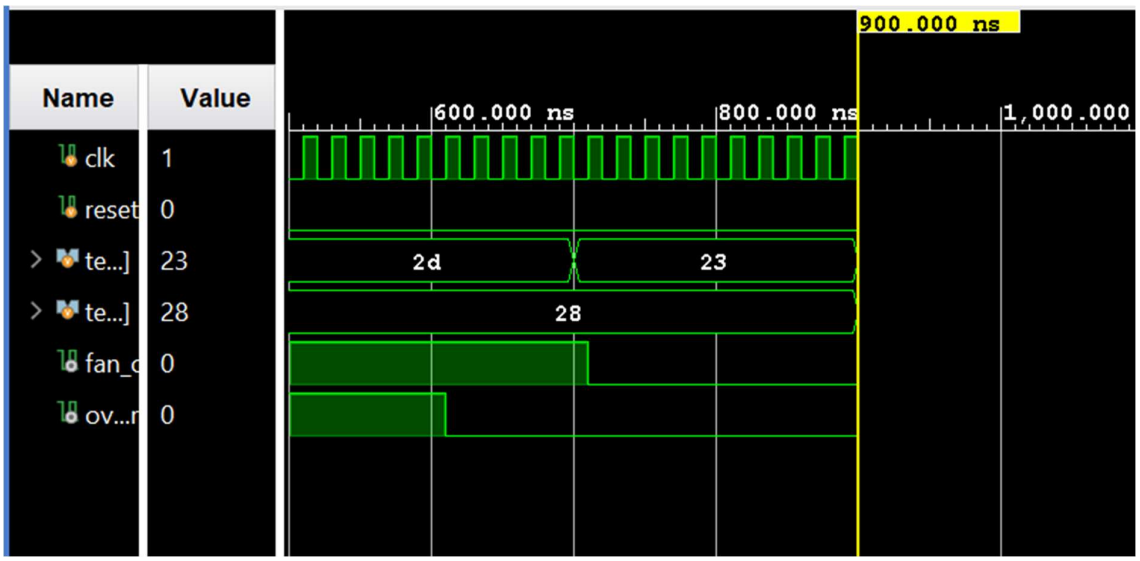


Figure 9 Timing diagram of the temperature fan controller.

- Light controller simulation verifies lighting control based on ambient light

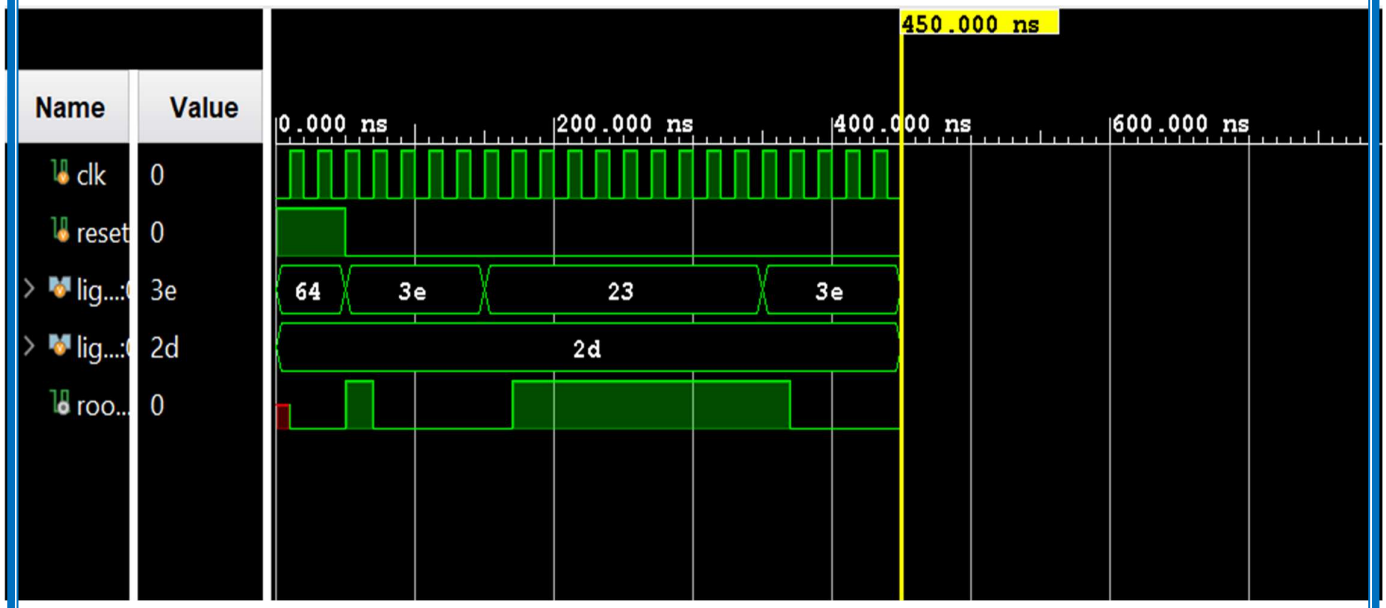


Figure 10 Timing diagram of the light controller

- Intruder alert controller simulation verifies FSM behaviour and alert generation

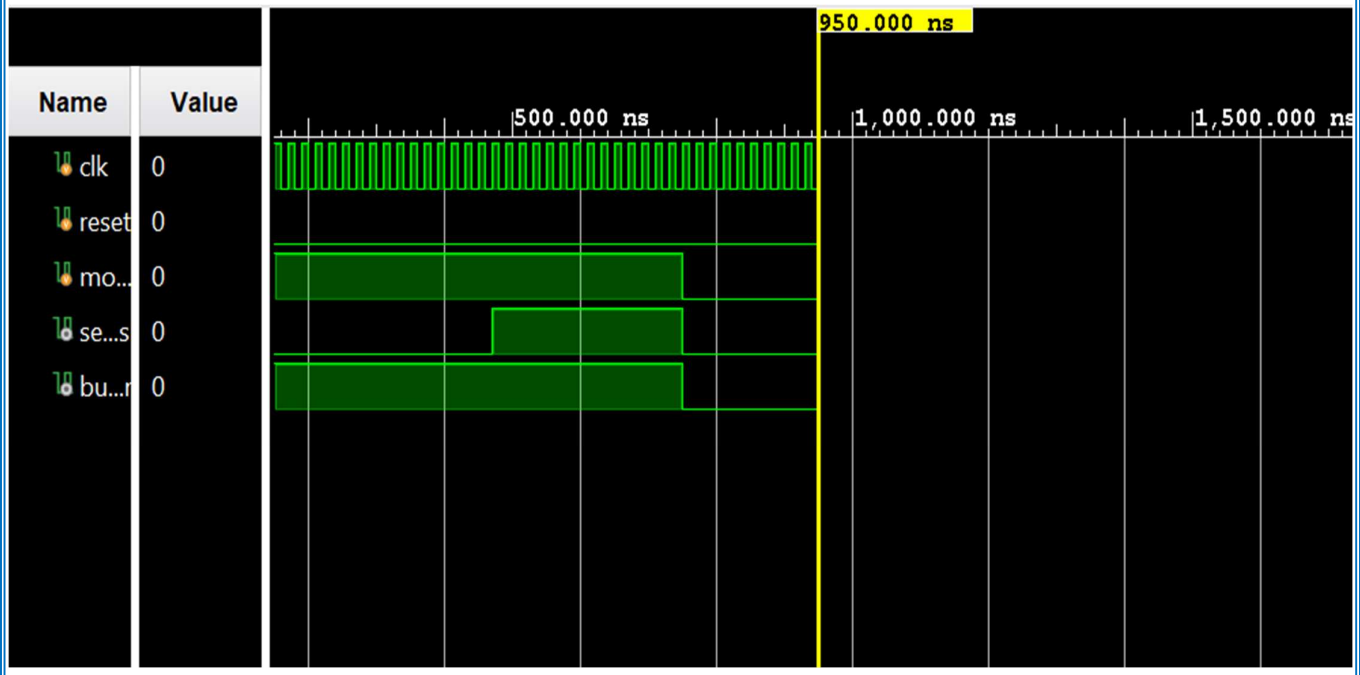


Figure 11 Timing diagram of the Intruder Alert Controller showing alert activation

- Top-level simulation confirms correct module integration

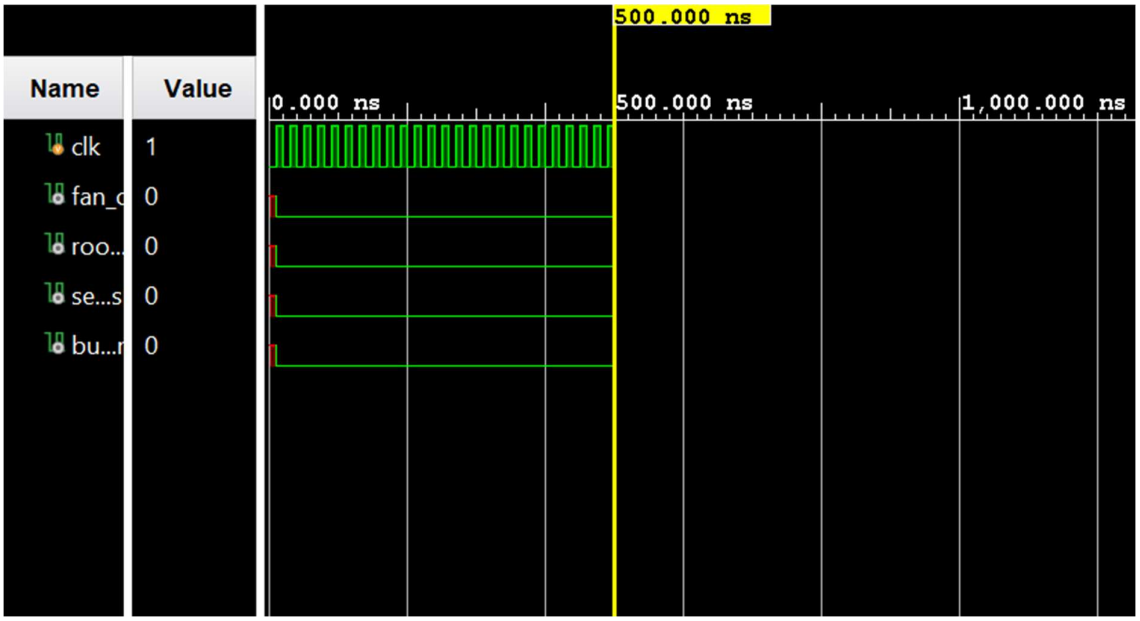


Figure 12 Behavioral simulation waveform of the top-level controller

7. FPGA Implementation

The complete design was synthesized and implemented using Xilinx Vivado targeting an Artix-7 FPGA (xc7a35tcpg236-1). Pin constraints were defined using an XDC file, and a valid bitstream was successfully generated.

A reduced I/O wrapper was used only for bitstream generation to simplify pin constraints, while the original architectural top-level controller is presented for documentation and simulation.

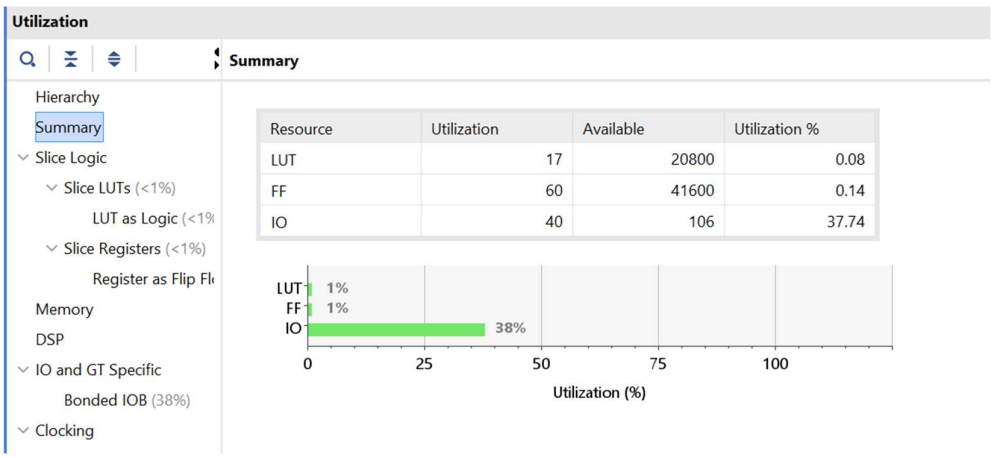


Figure 13 FPGA resource utilization summary

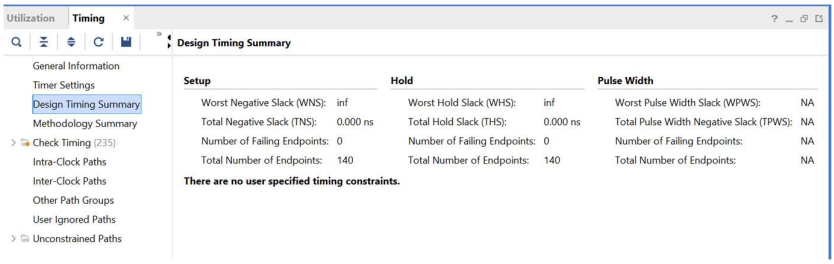


Figure 14 Timing analysis summary after synthesis

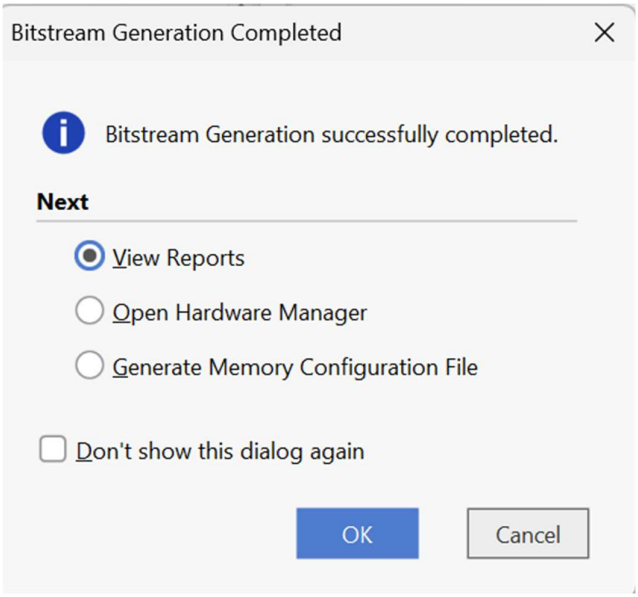


Figure 15 Successful bitstream generation in Vivado

Signal	Direction	FPGA Pin	I/O	Description
clk	Input	W5	LVCMOS33	System clock input
fan_on	Output	U16	LVCMOS33	Fan control signal
room_lights	Output	V16	LVCMOS33	Room lighting control
security_leds	Output	U14	LVCMOS33	Security LED indicator
buzzer_on	Output	V14	LVCMOS33	Intruder alert buzzer

Figure 16 FPGA pin mapping defined using Xilinx Design Constraints

8. Resource Utilization and Timing Analysis

As shown in figure 13 and 14 the synthesis results show that the design uses minimal FPGA resources, indicating an efficient implementation. Timing analysis confirms that there are no timing violations, and the design meets FPGA timing requirements.

9. Conclusion

In this project, an FPGA-based Smart Home Controller was successfully designed, simulated, and implemented. The modular architecture allowed easy integration of multiple control functions such as temperature regulation, lighting control, and intruder detection. The use of FPGA technology ensures fast response and reliable operation.

The project demonstrates the practical application of Verilog HDL and FPGA design techniques in real-world automation systems.

10. Future Scope

Future enhancements of this project may include:

- Integration of wireless communication modules
- Support for mobile app-based control
- Sensor interfacing with real hardware
- Power optimization techniques
- Expansion to control additional home appliances